Test Plan

Author: <Author Name>

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Agenda

- **Duration:** *2 Hours*
- **Audience:** *Testers*
- **Content:**
  1. Test plan process
  2. Test plan structure
  3. Quiz

Objectives

After the course, student will:

- Understand the way to create a Test Plan:
  - Input
  - Output
- Using Test Plan checklist to review a Test Plan
- Know Test tools available to use

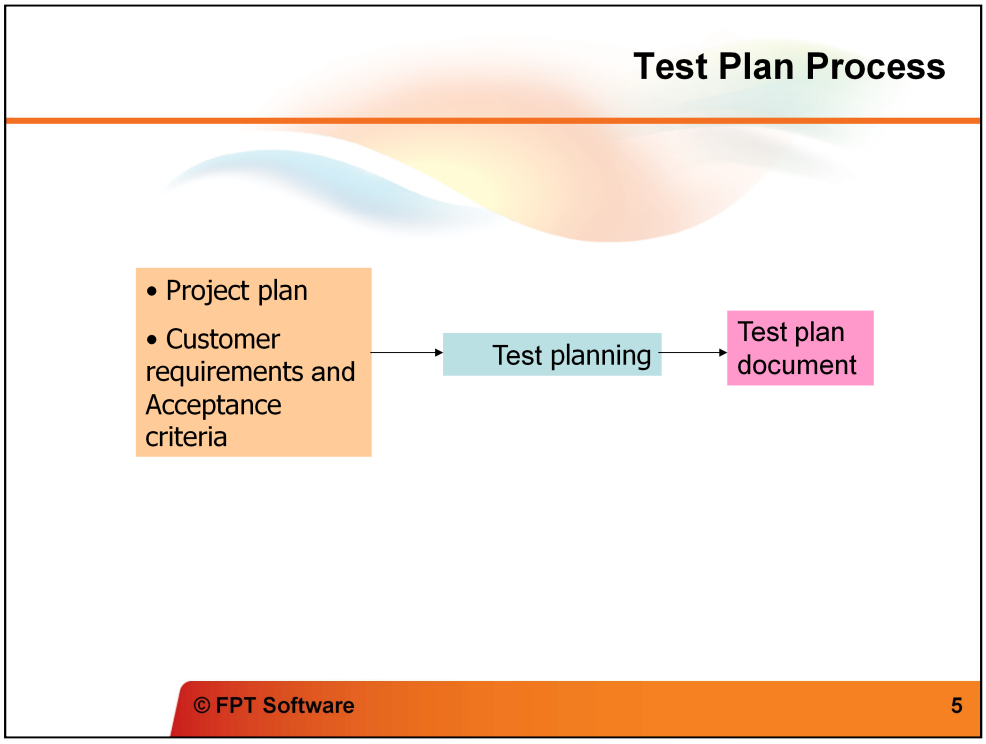
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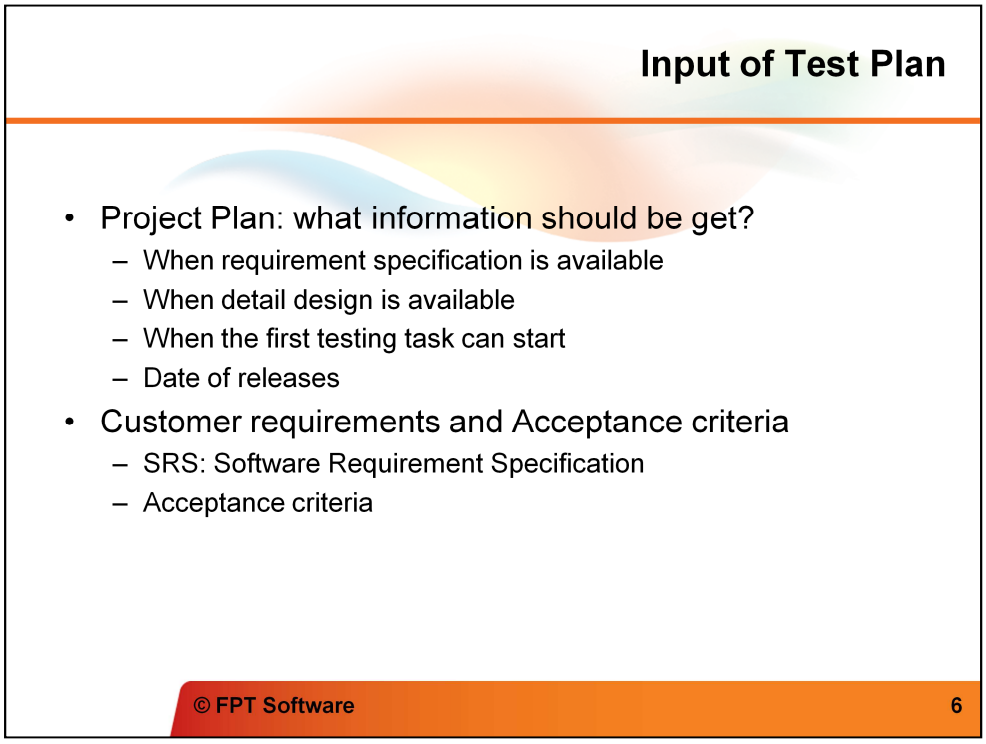
Test Plan Objective

- Identify testing scope
- Identify testing risks
- Identify acceptance criteria
- Obtain commitment from affected groups
- Brings in Effective collaboration among the teams
- Specify deliverables
- Define test strategy
- Define the Responsibilities
- Decide on Automation Early
- Decide on Test Metrics & Manage through Metrics

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This slide is used to show the input and output of creating test plan



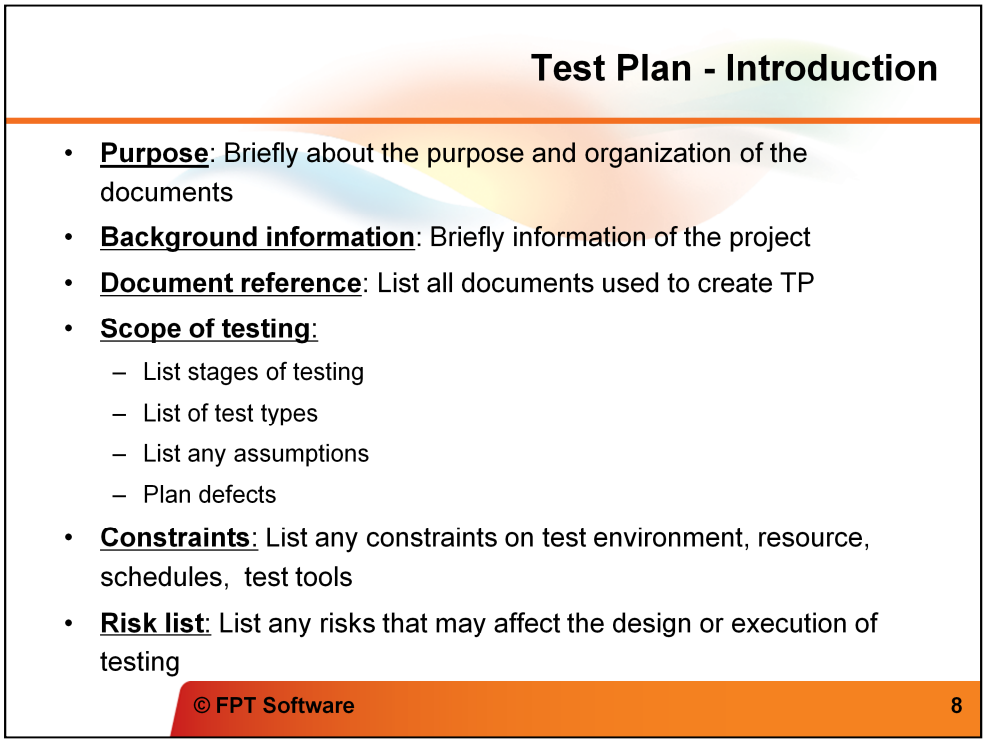
Give student example and help student to read & understand Project Plan and SRS.

Test Plan Structure

1. Introduction
2. Define requirements for test base on acceptance criteria
3. Define test strategy base on requirements for test: test types, stage, tools
4. Define resources and responsibilities
   - Define the system: hardware & software for testing
5. Define Test milestones
6. Define deliverables of test: TP, TC, TR

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+ TP = Test Plan
+ TC = Test Case
+ TR = Test Report

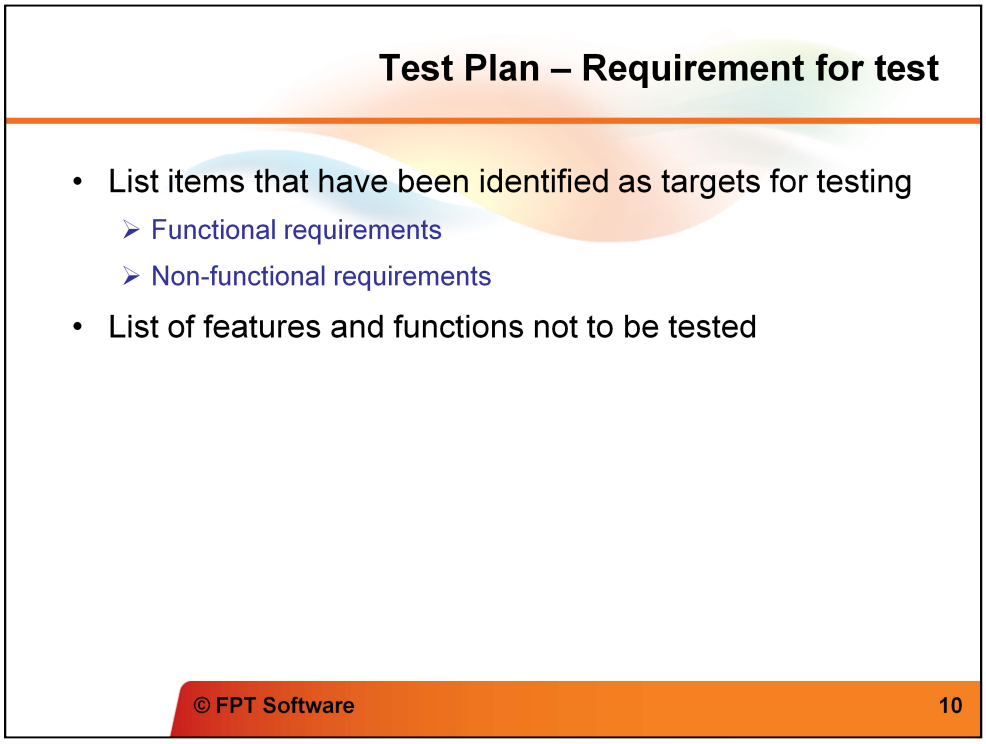


1. Example of risk:
   - Risks in a Computer Environment: Inability to control technology, Incorrect Entry of Data
   - Risks that Impact Testing: Number of Qualified Resources, Test Environment, Tools and Procedures, New Technology

2. Note: If testers find new risk (affect test) that hasn't defined in project plan yet then they must inform PL to update Project Plan. (PL = Project Leader)

Test Stages - Types

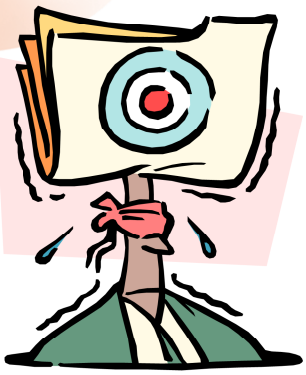
- **Test Stage:** The stages in which the test will be executed
  - Unit test
  - Integration test
  - System test
  - Acceptance test
- **Test Types:** all Types of test to execute:
  - Function test
  - User interface test
  - Performance test
  - Security and Access Control Testing
  - Regression Testing
  - ….



Divide mainly to 2 type: functional requirements and non-functional requirement

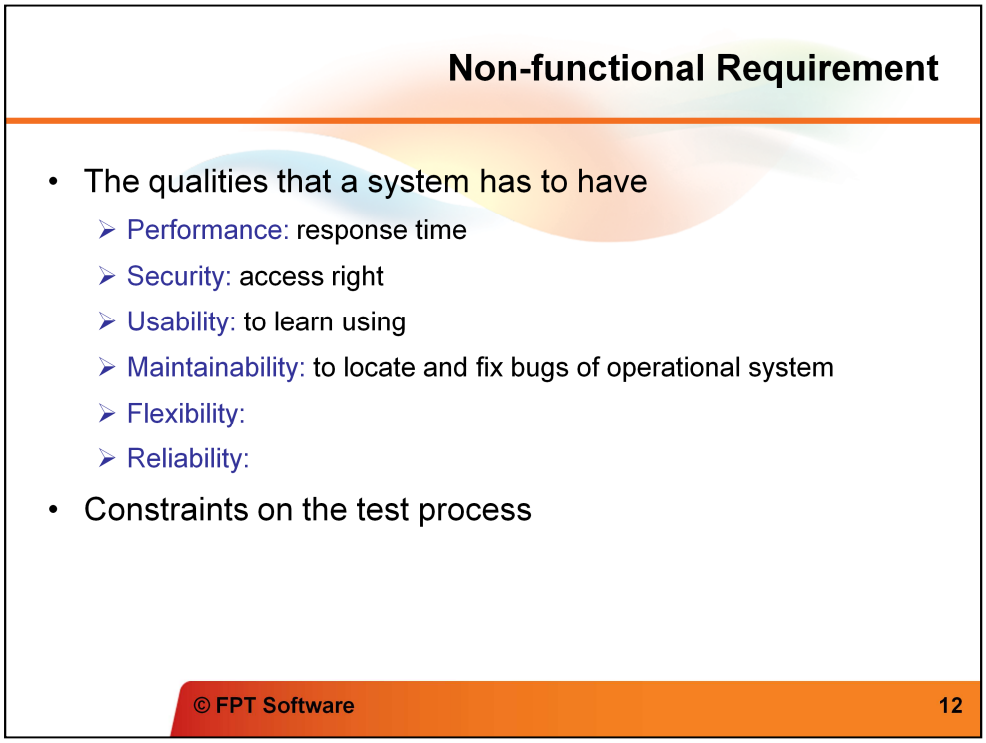
Functional Requirement

- Things that a system has to do related to business flow
- Example:
  - Do a calculation: add, new, edit, delete functions
  - Make a decision: check user right
  - Make a report



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Make a decision: a kind of security test



Example:

1. Performance requirement: response time of a specific function or module
2. Security:
    a. Protect from out-site (Ex: IMJ- SRS-requirement: Do following methods to against crack from out site)
        - ✓ 1. Buffer Overflow
        - ✓ 2. Cross Site Scripting
        - ✓ 3. Parameter Manipulation
        - ✓ 4. Backdoor & Debug Options
        - ✓ 5. Forceful Browsing
        - ✓ 6. Session Hijacking／Replay
        - ✓ 7. Path Traversal
        - ✓ 8. SQL Injection
        - ✓ 9. OS Command Injection
        - ✓ 10. Client Side Comment
        - ✓ 11. Error Codes）
    b. Decode private information
    c. Database security requirement: ex: system must do not contain the function, service that will not be used, account access, password setup, etc
3. Usability: (PT_LMS project requirement)

Test Strategy: General

- **Objective:** what needs to be ensured
- **Technique:** how testing will be executed
  - What will be tested
  - The major actions to be taken during test execution
  - The methods used to evaluate the results
- **Completion Criteria:**
  - Identify acceptance criteria for product quality
  - Identify when the testing is successfully executed
- **Special Considerations:**
  - Identify any influences/dependencies which may impact/ influence the testing described in Test strategy.
  - Criteria to stop testing (UT not good/meet requested Test coverage/…)

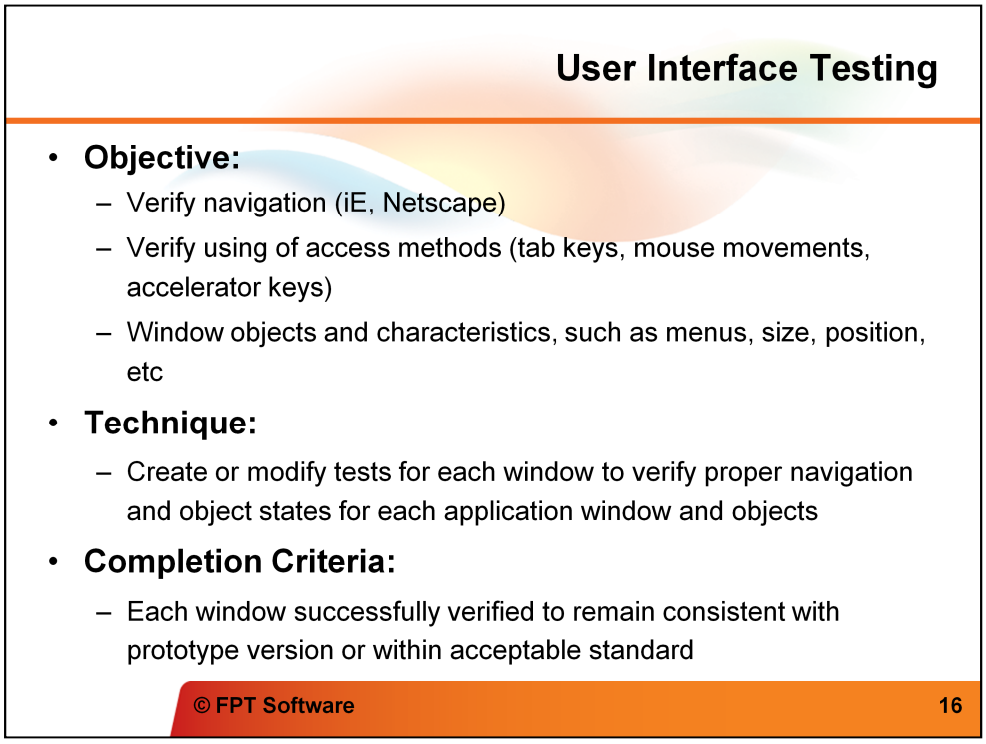
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Function Testing

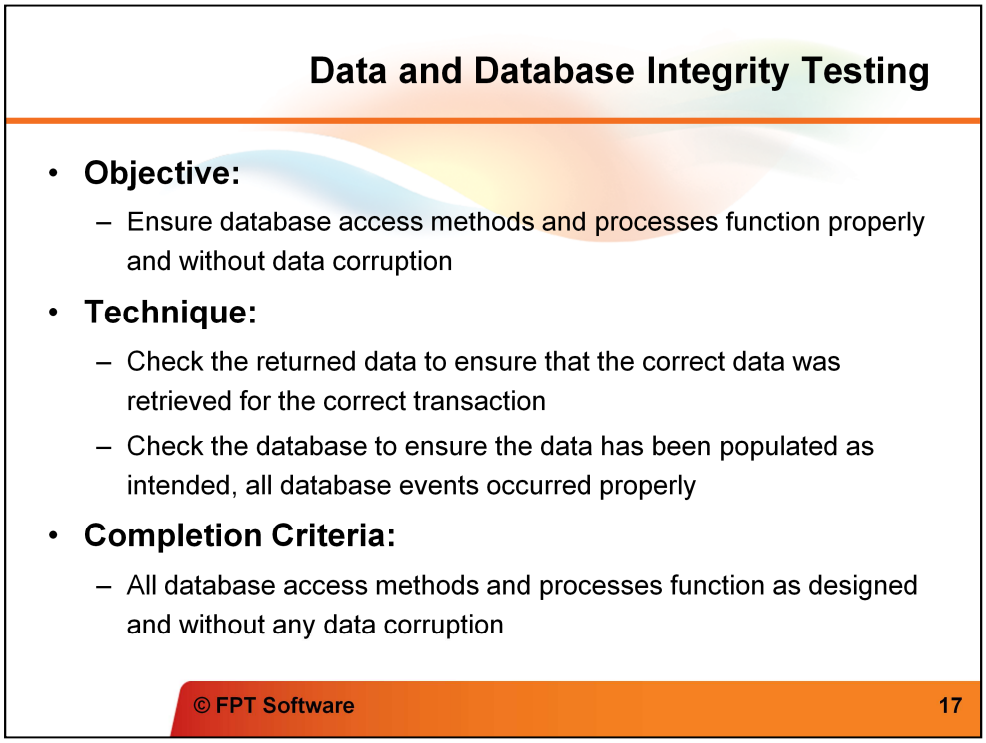
- **Objective:** Ensure proper target-of-test functionality, including navigation, data entry, processing, and retrieval
- **Technique:** based on black box techniques
  - Verifying the application and its internal processes by interacting with the application via the Graphical User Interface (GUI)
  - Analyzing the output or results
- **Completion Criteria:**
  - All planned tests have been executed
  - All identified defects have been addressed

Function Testing (cont…)

- **Special Considerations:** Identify or describe those items or issues (internal or external) that impact the implementation and execution of function test:
  - Test database
  - Criteria to stop testing (UT not good/met requested Test coverage/…)



We have a guideline to write test case for GUI. More detail in TC trainning



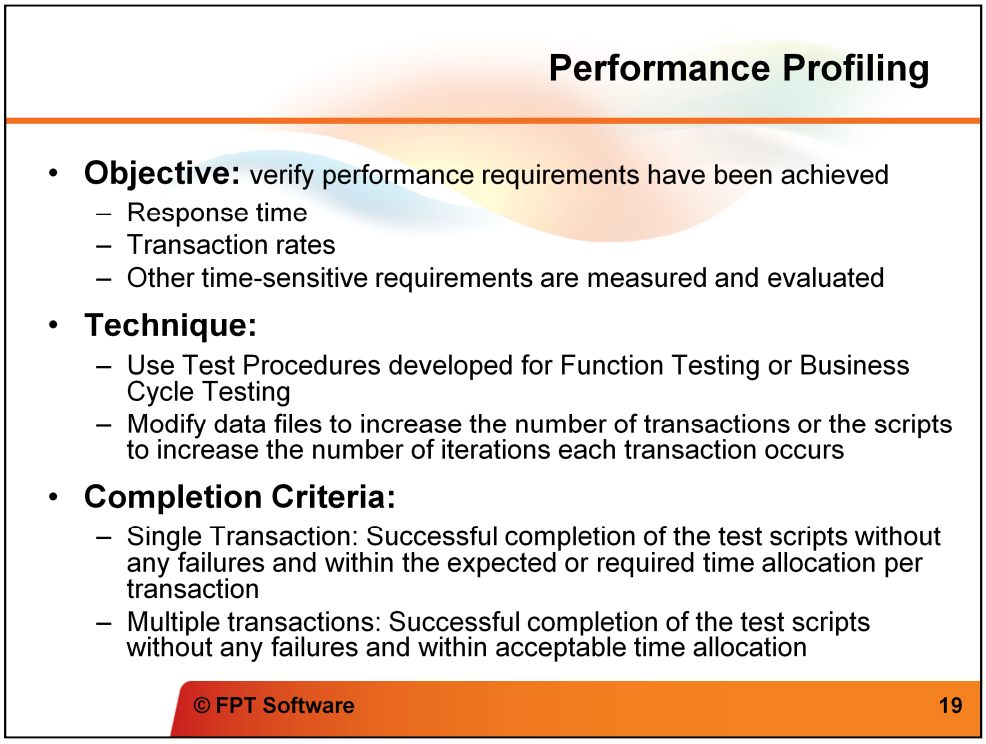
Whenever there is a requirement about integrating data and database so this test type is using

Ex: When your project have a requirement that you have to integrate your database with data from customer so this test type is using

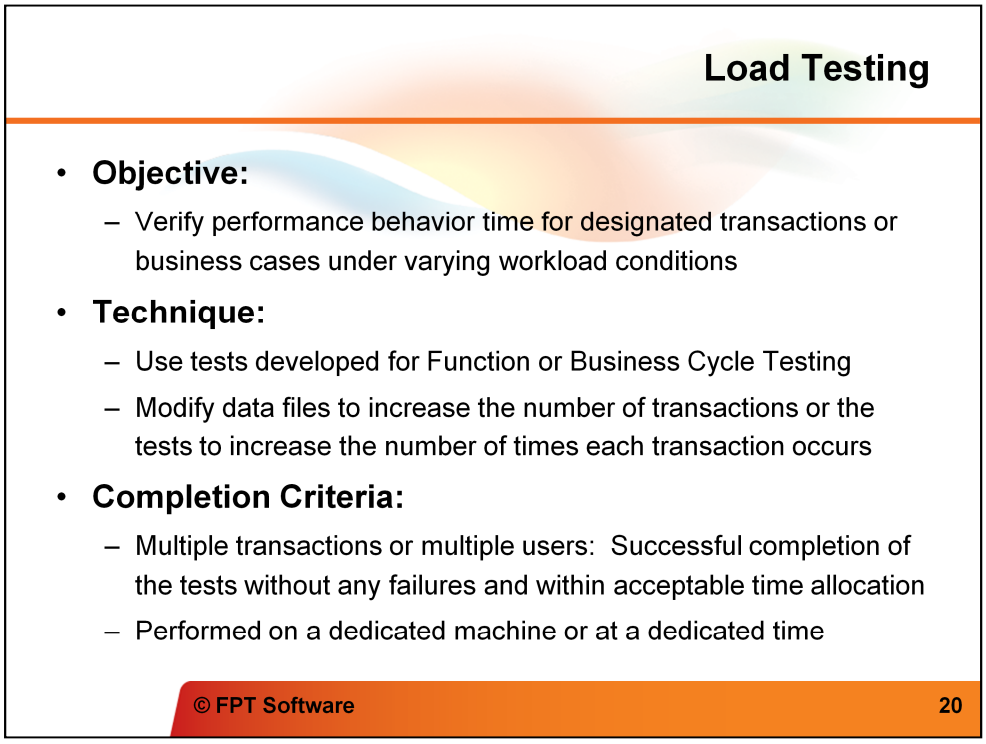
Technique: actually to ensure that the integration is successful or not, testers must regression test all the system base on written test cases

Business Cycle Testing

- **Objective:**
  - Ensure proper target-of-test and background processes function according to required business models and schedules
- **Technique:**
  - All functions that occur on a periodic schedule will be executed or launched at the appropriate time
  - Testing will include using valid and invalid data
  - Each business rule is properly applied



Check response time of a single transaction or multi transaction with provided CM



Ex: there is a requirement of specific number of concurrent users running

Stress Testing

- **Objective:** verify functions work without error under the following stress conditions:
  - Little or no memory available on the server (RAM)
  - Maximum actual or physically capable number of clients connected or simulated
  - Multiple users performing the same transactions against the same data or accounts
- **Technique:**
  - Use tests developed for Performance Profiling or Load Testing
  - To test limited resources, tests should be run on a single machine, and RAM on server should be reduced or limited
- **Completion Criteria:** Get the specific the limited resources, the min, max number of transactions or concurrent users that make application run fail

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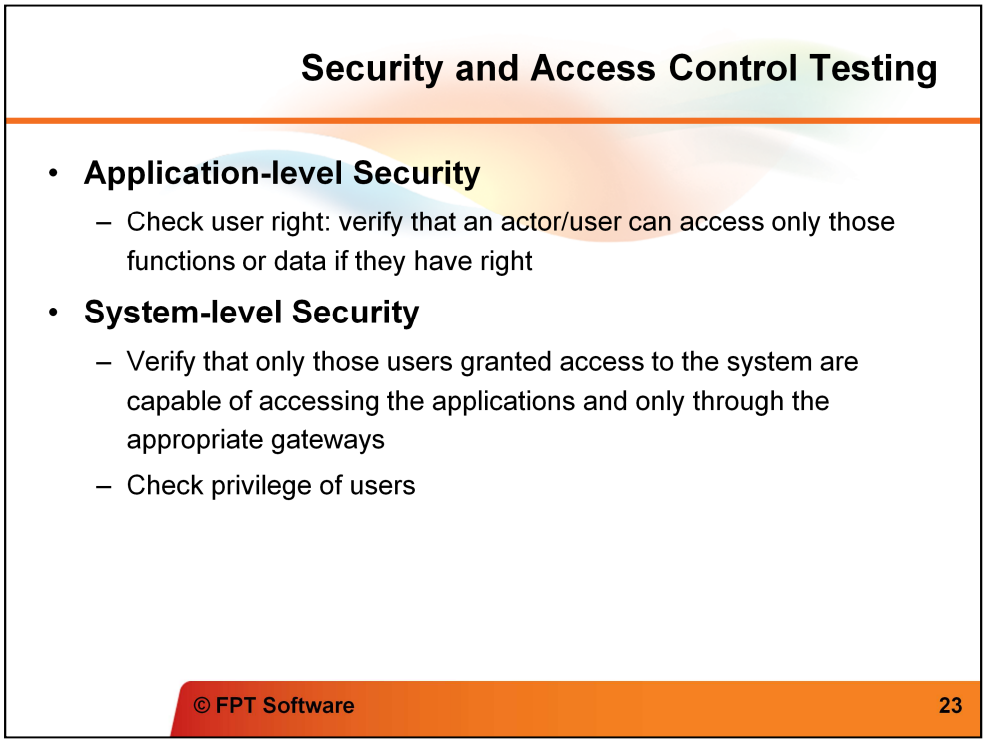
Ex: there is a requirement to

-Define the min, max concurrent users can perform a function

-Define the min configuration that system can run, ect

Volume Testing

- **Objective:**
  - Test with large amounts of data to determine if limits are reached that cause the software to fail
- **Technique:**
  - Data and database integration
  - Insert data (by manual/tools/scripts)



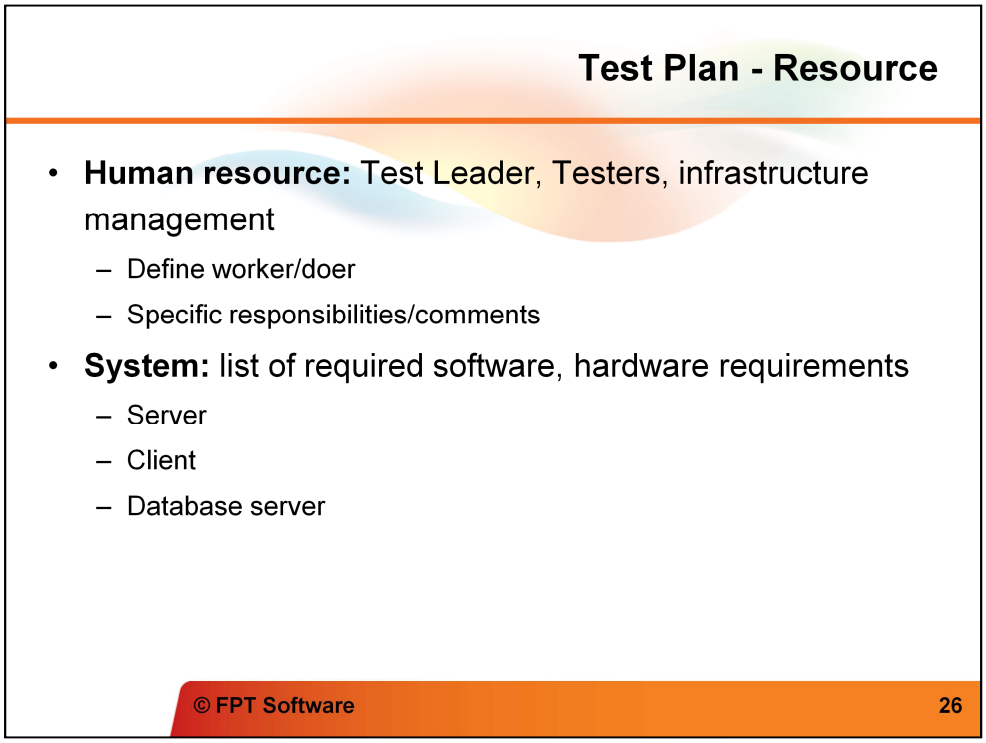
Example: one user cannot log in system from 2 machines

Regression Testing

- **Objective:**
  - Validate modified parts of the software, to make sure that the modification does not cause errors in other parts
- **Technique:**
  - Reuse the set of test cases from an existing test suite to test a modified module

Test Tool

- **List test tools will be employed for this project**
  - Tool for log defects/bugs
  - Load test tool
  - Performance test tool
  - Automatic generated data tool
  - Tools are used for Graphic test
- **Common Test tools:**
  - Eye Dropper: for colors
  - VRuler: to measure the distances
  - Unit Test tools: MAULA, JUnit, NUnit, …
  - Function test tools: Rational Robot test, Ruby-Watir
  - Load test: Rational Load test, OpenSTA



Note: testing environment must be listed and if the real testing environment is different with the required testing environment then it should be listed:

-Testing environment as customer's require (software and hardware)

-The real testing environment

Test Plan - Estimation

- **Two techniques for estimation covered by ISTQB:**
  - Metrics-based: involves analyzing metrics from past projects and from industry data
  - Expert-based: involves consulting the people who will do the work and other people with expertise on the tasks to be done

Metric-based Estimation

- Can be as simple or sophisticated as you make it
- The simplest approach is to ask:
  - How many testers do we typically have per developer on a project?
  - Classifying the project in terms of size (small, medium or large) and complexity (simple, moderate or complex) and then seeing on average how long projects of a particular size and complexity combination have taken in the past
- Another approach: look at the average effort per test case in similar past projects and to use the estimated number of test cases to estimate the total effort

Expert-based Estimation

- Asking the individual contributors and experts involves working with experienced staff members to develop a work-breakdown structure for the project
- Called 'bottom up' estimation because you start at the lowest level of the hierarchical breakdown in the work-breakdown structure - the task - and let the duration, effort, dependencies and resources for each task add up across all the tasks

Test Estimation - Reality

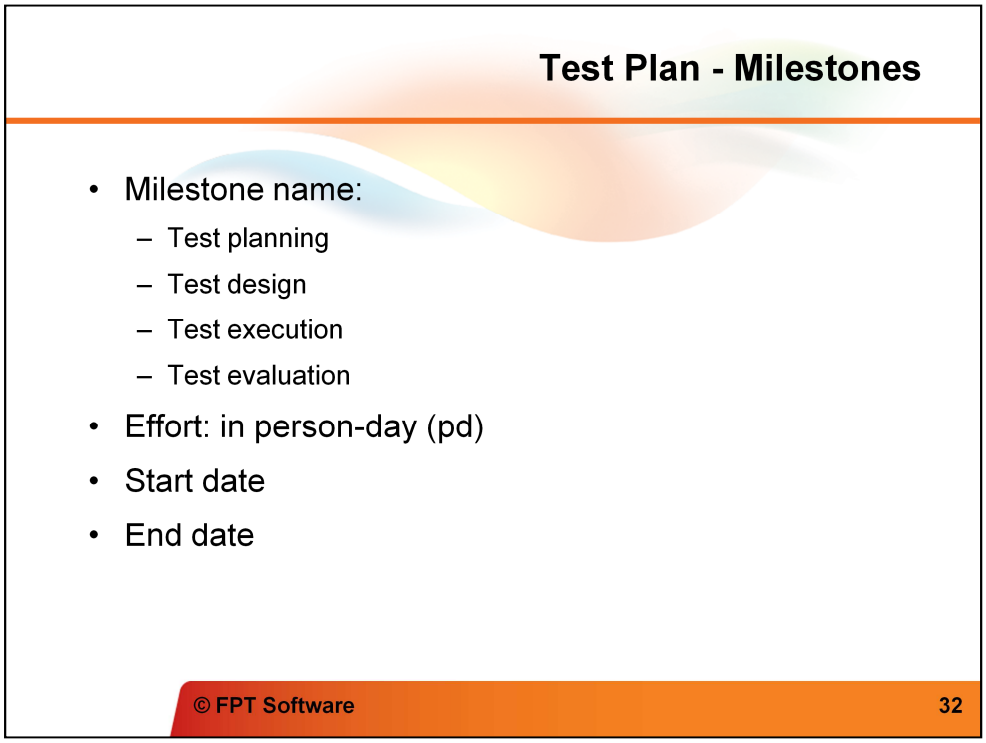
- **Work Break Down:**
  - List all requirement for test (functional and non functional)
  - Divide requirement for test into 3 category of complexity: simple, average, complex
  - For each category, input estimated effort for test tasks: study requirement, create test case, test data, etc.
  - Add effort for test plan and other management works (if have)

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Based on ISTQB book

Test Estimation - Reality

- **Historical Data:**
  - Based on effort rate of testing/project size
  - Based on domain
  - Based on market: different market -> different rates
  - Got total effort for test
  - Estimate for each tasks of test milestone



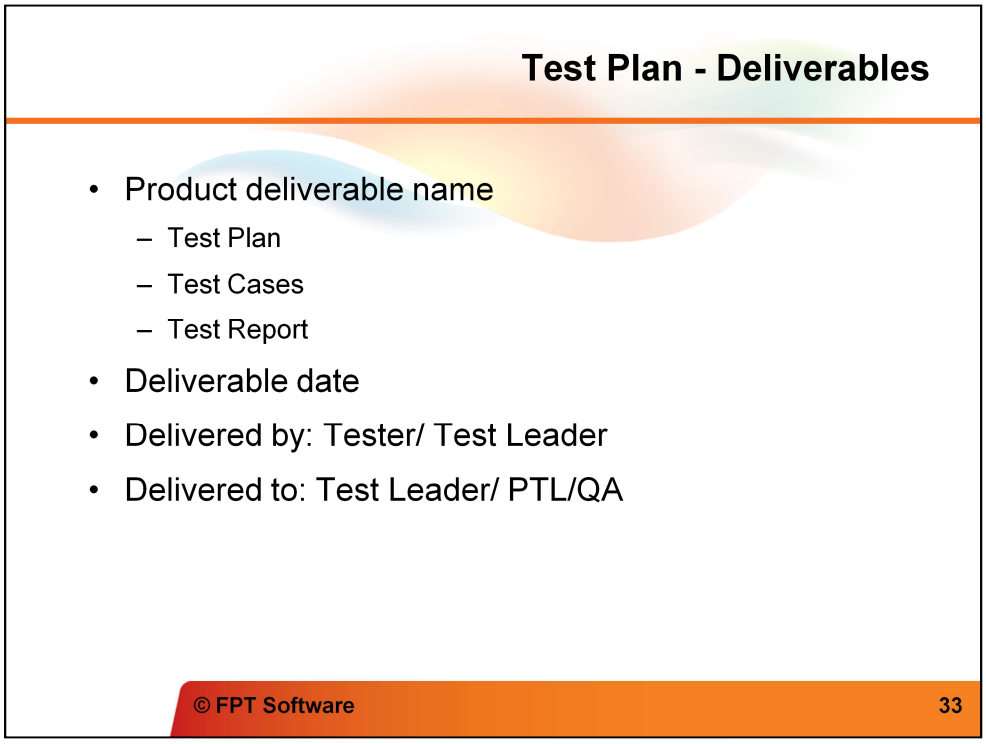
Milestone name: base on commitment with customer (the document which will be released) and Company's process (TP, TC, etc)

Effort: base on the fact to estimate, do not only use the effort that PL estimate for milestone.

Do not base on calendar effort to apply the effort for each milestone, ex: as plan in PP, we can start creating TP from 10.Dec.05 to 12.Dec.05, some people set effort of creating TP is 2MD and start date is 10.Dec.05 and end date is 12.Dec.05

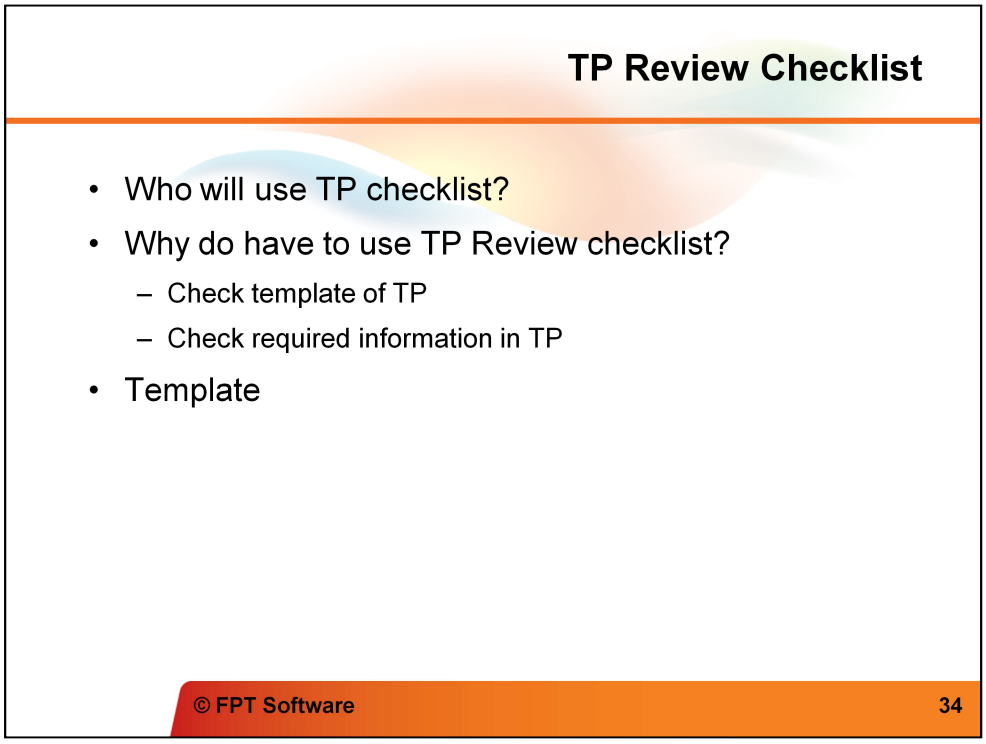
Start date: Not only base on the project plan but also base on the process, ex: when Project plan and SRS finish then tester can create TP so the start date of creating TP can be right after the day that got PP and SRS

End date: If we have to Project plan, one for customer and one for internal then we should keep the end date of each test milestone the same as internal project plan.



PTL = Project Team Leader or Project Technique Leader

QA = Quality Assurance


TP Review Checklist
• Who will use TP checklist?
• Why do have to use TP Review checklist?
– Check template of TP
– Check required information in TP
• Template
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TP = Test Plan

Resources & references

- **Resources**
  - Template_Test Plan.doc
  - Checklist_Test plan review.xls
- **References:**
  - Foundations of Software Testing.pdf
  - Guideline_Software Testing.pdf
  - Process_Test.pdf

QUESTIONS AND ANSWERS